

BEFAST AI — Conclusion

Multimodal AI-assisted screening for the BE-FAST stroke warning signs — feasibility, evaluation methodology and limitations

1. Conclusion

The integration of the individual BEFAST components allows the system to perform a more comprehensive preliminary assessment than relying on a single symptom. Information obtained from Balance, Eyes, Face, Arms and Speech, together with symptom onset time, is combined to produce an overall screening result. The combination rule is deliberately conservative: a single sign judged present, or merely uncertain, is sufficient to escalate the whole assessment and open the emergency pathway, so the system is biased toward false alarms rather than missed detections.

The prototype demonstrates the feasibility of integrating multiple forms of data — video frames, still images, audio and reported symptom information — into a single AI-assisted BEFAST screening application. The assistance is twofold: pretrained machine-learning models supply the low-level measurements, in the form of facial and postural landmarks and a speech transcript, while an explicit rule layer converts those measurements into a per-sign judgement. This multimodal approach may be useful because stroke can present with different combinations of symptoms among individuals, and because a sign that one modality cannot capture may still be reported through the observer questionnaire, which overrides the automated result when the two disagree.

Overall, the project demonstrates the feasibility of combining the BEFAST principle with artificial intelligence to create an accessible preliminary stroke-screening application that runs entirely in a web browser without an application server. Further development should focus on assembling a labelled dataset, validating the system against clinically appropriate reference data, calibrating the decision thresholds to improve sensitivity and specificity, and evaluating the application under realistic conditions. With appropriate clinical validation, the BEFAST AI concept may contribute to faster recognition of suspected stroke and encourage timely access to emergency medical care.

2. Present validation status

No accuracy has yet been established, and none is claimed. The fifteen decision thresholds were selected by inspection and are declared unvalidated in the source; there is no labelled dataset, no clinical reference standard and no automated test suite. The application therefore reports categorical urgency tiers rather than a probability, and labels every displayed figure a prototype estimate. The thresholds are also not the complete decision surface: feature weights, the ten-second hold, the 2.2-second target exposure and the eight-second recording window are undocumented constants inside the modules.

3. Recommended evaluation methodology

Reference standard. Ground truth should be the discharge diagnosis established by neuroimaging and neurologist assessment; a lower-cost initial study may substitute concurrent NIHSS or FAST-ED scoring by a certified examiner. Because each sign is judged separately, labels are required per sign and not only per participant.

Study design. A prospective paired-comparison study, with the application and the reference standard blinded to each other. Recruitment should include stroke mimics such as Bell's palsy, migraine, seizure and hypoglycaemia alongside healthy controls, since specificity estimated from a stroke-only cohort is uninformative.

Primary metrics. Sensitivity, specificity, PPV and NPV with 95% confidence intervals, reported per sign and for the screen as a whole. For an application whose output is a recommendation to call emergency services, sensitivity is the quantity to prioritise, as a missed detection is the more harmful error.

Threshold calibration. The continuous features should be swept across ROC curves, with AUC reported and operating points fixed at a pre-specified minimum sensitivity rather than chosen by inspection. Three output tiers imply two thresholds per sign, so an explicit misclassification-cost criterion is needed.

Reliability. Cohen's kappa against clinician judgement and between observers completing the questionnaire; Bland-Altman comparison of the continuous measures against manual annotation; test-retest repeatability; and inter-device variance across handsets, lighting and camera distance, since landmark quality depends on all three. Subgroup analysis by skin tone, age, eyewear, facial hair, seated posture and language is also required.

4. Measurement issues to resolve first

Review of the feature extraction identified issues that would bias a validation study before recruitment. Arm drift is divided by an arm length recomputed on the final frame, so as the arm falls the divisor shrinks and the normalised drift inflates; only the first and last frames contribute, and the comparison is signed, so upward drift cannot register. Postural sway is a standard deviation normalised by the last frame's shoulder width, which dilutes a single large lurch. The eyelid and gaze figure is taken from whichever animation frame happens to be last, so an ordinary blink can determine the result. Transcript accuracy is order-insensitive word matching, so a scrambled utterance of the correct words scores full marks. Separately, each measurement is stored as a formatted display string with the underlying number discarded, which prevents any later recalibration; numeric feature logging, a consented export path and an offline replay harness are prerequisites for the study above.

5. Limitations

A single camera yields two-dimensional landmarks that only approximate three-dimensional motion, and quality degrades with poor lighting, oblique angles and distance. Camera frames are never transmitted, but speech transcription is performed by a browser service that sends audio to the vendor, and the machine-learning runtime and model weights are retrieved from public content delivery networks on first use, so the application requires network access and cannot operate offline. Speech recognition is unavailable in Firefox and inconsistent on iOS Safari, in which case the sign falls back to the questionnaire; the target phrase is English with the recogniser fixed to en-US and would score healthy Thai speakers as impaired. BEFAST does not capture every posterior-circulation presentation. The application is a screening aid and not a diagnostic device.